

EVENT MANAGEMENT SYSTEM

The Development of a Cross-Platform Event Management System Mobile Application using Flutter and Google Firebase Firestore Database

Objective

The primary objectives of this project are multifaceted, aiming to deliver a robust and user-centric mobile application for event management.

The Key goals include:

1. **Platform Development:** To engineer a fully functional, cross-platform mobile application compatible with both iOS and Android operating systems using the Flutter framework, ensuring a consistent and native-level user experience.
2. **Backend Integration:** To implement a secure, scalable, and real-time backend infrastructure utilizing Google Firebase services, specifically:
 - **Firebase Authentication:** For robust user identity management, supporting both email/password and Google Sign-In authentication methods.
 - **Cloud Firestore:** To serve as the primary NoSQL database for storing, retrieving, and synchronizing event data in real-time across all client applications.
 - **Firebase Storage:** For the efficient and secure storage and retrieval of high-resolution event banner images and other media assets.
3. **Core Functionality:** To provide a comprehensive suite of features that enables users to:
 - Seamlessly create new events with detailed descriptions, dates, locations, and visual media.
 - Browse a publicly accessible, chronologically sorted feed of all upcoming events.
 - View in-depth details for any event of interest.
 - Manage their own created events.
4. **State Management & Architecture:** To employ a clean and maintainable software architecture using the Provider package for efficient state management, ensuring the application is scalable and easy to debug and extend.

Introduction

➤ Overview:

This Flutter-based application enables users to browse a public feed of upcoming events, create detailed event listings with rich media, and manage their event portfolio with full editing capabilities.

The system eliminates the traditional hassles of event promotion and discovery by offering a dedicated digital space where events are organized, searchable, and accessible in real-time.

Whether it's a local community gathering, a professional workshop, or a large-scale conference, the Event Management System streamlines the entire event lifecycle from conception to execution.

➤ Key Features:

1. User Authentication:

- Secure login via Email/Password

2. Event Creation & Management:

- Intuitive interface for creating and browsing multiple events

3. Real-time Event Feed:

- Public browsing of all upcoming events with live updates

4. Media Integration:

- Support for event images with cloud storage

5. Cross-platform Compatibility:

- Native performance on both iOS and Android devices

➤ Technology Stack:

1. Frontend:

- Flutter framework with Dart programming language

2. Backend:

- Google Firebase ecosystem

3. Database:

- Cloud Firestore Database(No SQL) for real-time data synchronization

4. Authentication:

- Firebase Authentication with multiple provider support

5. State Management:

- Provider package for efficient app state handling

➤ Target Audience:

1. Primary Target Groups:

- Event Organizers & Planners, Educational Institutions, Corporate Event Managers, Community Organizers And Small Business Owners.

2. General Users & Attendees:

- Socially Active Individuals, Professional Networkers, Hobby Enthusiasts, Students & Young Adults, Tourists & New Residents

3. Secondary Target Groups:

- Educational & Institutional Users, Business & Commercial Users

Problem Statement

Currently, The Challenges in Event Management is:

1. Fragmented Information Distribution:

- Event details scattered across multiple platforms (social media, emails, physical posters).
- No centralized repository for event discovery.
- Difficulty in finding comprehensive event details in one place.

2. Inefficient Promotion & Discovery:

- Organizers struggle with limited reach and visibility.
- Attendees face difficulty discovering relevant local events.
- High effort required to find quality, trustworthy events.

3. Management Inefficiencies:

- Manual processes for event creation.
- Difficulty in managing multiple events simultaneously.

4. Technical Limitations:

- No mobile-optimized solutions for event management.
- No cloud-based backup for event data.

5. Accessibility Issues:

- Language and cultural barriers in event information.
- Time zone challenges for international events.

6. Trust & Reliability Concerns:

- Difficulty in verifying event authenticity.
- Risk of fraudulent events and scams.

7. Cost Inefficiencies:

- High marketing costs for event promotion.
- Manual processes increasing operational costs.
- No cost-effective solutions for small organizers.
- Hidden costs in traditional event management methods.

➤ Impact of These Problems:

■ For Organizers:

- Reduced event visibility and attendance.
- Increased time and resource expenditure.

■ For Attendees:

- Missed opportunities for relevant events.
- Wasted time and money on unsuitable events.
- Lack of trust in event quality and authenticity.

■ Overall Market Impact:

- Reduced community engagement.
- Inefficient utilization of event spaces and resources.

Software Requirement Specification(SRS) in IEEE format

➤ **Purpose:**

This document specifies the requirements for the Event Management System mobile application. It serves as a formal agreement between developers and stakeholders, detailing functional and non-functional requirements to guide development, testing, and validation.

➤ **Document Conventions:**

- FR: Functional Requirement
- NFR: Non-Functional Requirement
- Requirements are numbered hierarchically (e.g., FR-1.1, NFR-2.3)

➤ **Intended Audience:**

- Project Managers, Development Team, Quality Assurance Team, Client/Stakeholders And End Users.

➤ **Project Scope:**

The system shall be a mobile application for event management including user registration, event creation, event browsing, and user authentication. The system will NOT include event update/delete functionality, payment processing, or advanced analytics.

1. **Overall Description:**

● **Product Perspective:**

The system operates as a standalone mobile application integrated with Firebase services.

● **Component Diagram:**

User Device → Flutter App → Firebase → [Auth | Firestore | Storage]

● **User Classes:**

Class	Description	Key Permissions
Event Organizers	Authenticated users who create events	Create events, Upload images
Event Attendees	Authenticated users who browse events	Browse events, View event details
Guests	Unauthenticated users	Browse events only

- **Operating Environment:**
 - **Platform:** iOS and Android mobile devices
 - **Minimum OS:** Android 8.0 (API 26), iOS 12
 - **Network:** Requires active internet connection
 - **Storage:** Requires gallery access for image uploads
- **Assumptions:**
 1. Users have stable internet connection for core functionality
 2. Firebase services remain available and operational
 3. Users have basic smartphone literacy
 4. Event data does not require modification after creation

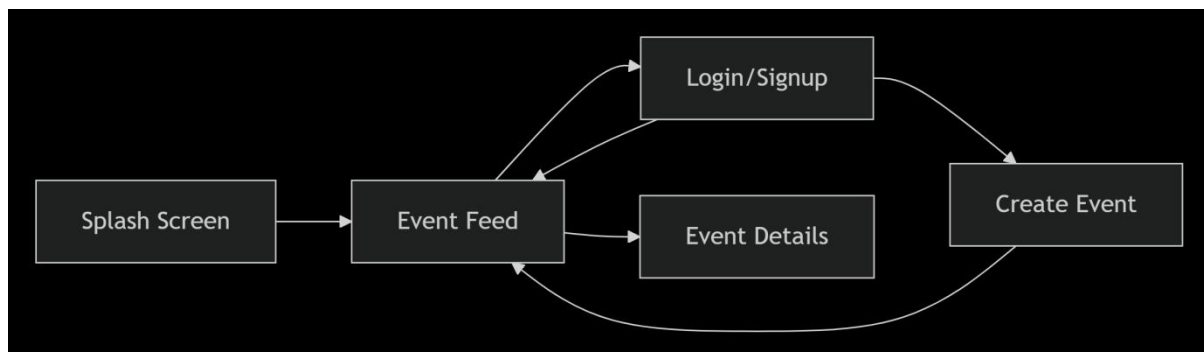
2. System Features:

- FR-1: User Authentication Module
- FR-2: Event Creation Module
- FR-3: Event Browsing Module
- FR-4: Media Handling Module

3. External Interface Requirements:

- **User Interfaces:**
 1. **Login Screen:** Email/password fields.
 2. **Event Feed:** Scrollable list of event cards with images and basic info
 3. **Event Creation Form:** Input fields for event details, image picker button
 4. **Event Details Screen:** Full event information display

➤ Screen Flow:



➤ Hardware Interfaces:

- Mobile device touches screen for user input
- Device storage for image selection
- Network interface for Firebase communication

4. Non-Functional Requirements:

- **Performance:**
 - 3-second event list load, 5-second authentication, 60fps smooth animations, Non-blocking image upload
- **Reliability:**
 - Handles network failures, No data loss, Data consistency maintained, 99% uptime
- **Usability:**
 - Intuitive interface, Consistent navigation, Accessible design, Clear error messages

- **Security:**
 - Secure login, Authorized access only, Data protection, No unauthorized access
- **Compatibility:**
 - Android 8.0+ & iOS 12+, All screen sizes, Portrait & landscape, Standard browsers
- **Maintainability:**
 - Clean code, Modular design, Best practices, Easy updates
- **Scalability:**
 - Auto-scaling, Efficient queries

➤ **Constraints:**

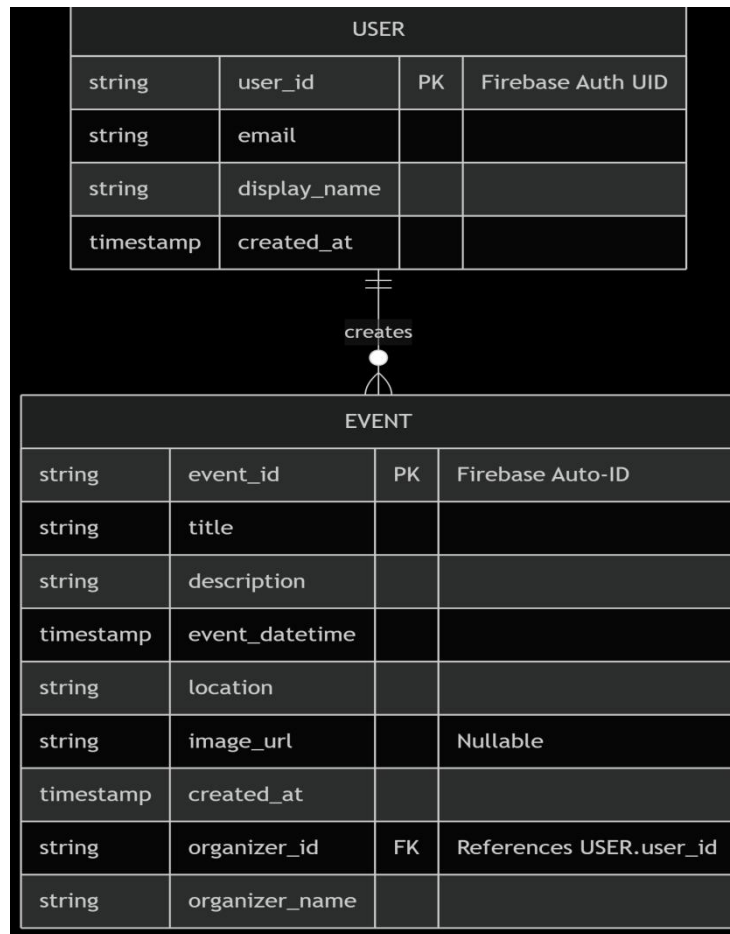
1. Event modification (update/delete) functionality is explicitly excluded from scope.
2. System depends entirely on Firebase services availability.
3. No offline functionality required.
4. Support for English language only in initial release.

➤ **Acceptance Criteria:**

1. All functional requirements shall be implemented and tested.
2. Application shall pass performance benchmarks on target devices.
3. User authentication shall work reliably across different networks.
4. Event creation and browsing shall function without data loss.
5. Application shall be deployable to Google Play Store and Apple App Store.

Entity Relation Diagram

➤ Visual ER Diagram (Simplified for NoSQL):



➤ Entities:

1. USER Entity

- **Primary Key:** user_id (Firebase Authentication UID)
- **Attributes:**
 - **email (String):** User's email address
 - **display_name (String):** User's display name
 - **created_at (Timestamp):** Account creation date
 - **Description:** Represents authenticated users in the system

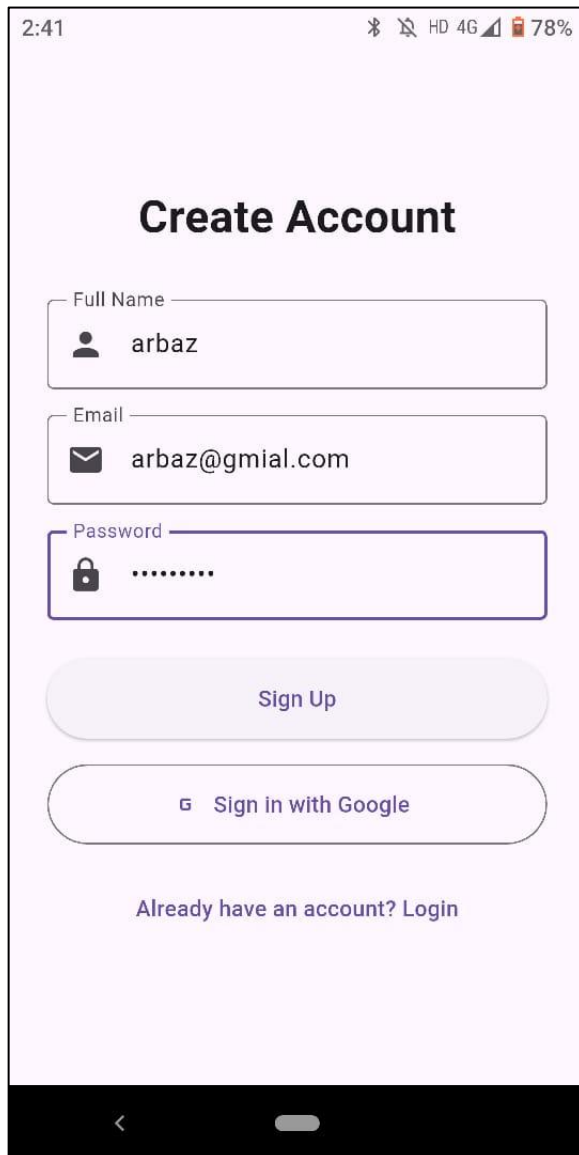
2. EVENT Entity

- **Primary Key:** event_id (Firebase Auto-generated Document ID)
- **Attributes:**
 - **title (String):** Event title (mandatory)
 - **description (String):** Event description (mandatory)
 - **event_datetime (Timestamp):** Event date and time (mandatory)
 - **location (String):** Event location (mandatory)
 - **image_url (String):** URL of event image from Firebase Storage (nullable)
 - **created_at (Timestamp):** Event creation timestamp
 - **organizer_id (Foreign Key):** References USER.user_id
 - **organizer_name (String):** Denormalized organizer name for quick access
- **Description:** Represents events created by users in the system

- **Relationships:**
 - **USER creates EVENT (One-to-Many Relationship)**
 - **Cardinality:** One USER can create zero or many EVENTS (1:N)
 - **Participation:** Partial participation for USER (some users may not create events), Total participation for EVENT (every event must have an organizer)
 - **Foreign Key:** organizer_id in EVENT table references user_id in USER table
- **Key Constraints:**
 1. **Primary Keys:**
 - **USER:** user_id (provided by Firebase Authentication)
 - **EVENT:** event_id (auto-generated by Firestore)
 2. **Foreign Key Constraint:**
 - Every organizer_id in EVENT must exist in USER's user_id
 3. **Referential Integrity:**
 - If a USER is deleted from Firebase Auth, their created events remain in the system (due to the constraint that events cannot be updated/deleted)

Snapshots of the different input and output screens

1. Sign-Up And Login:



2:41 HD 4G 78%

Create Account

Full Name

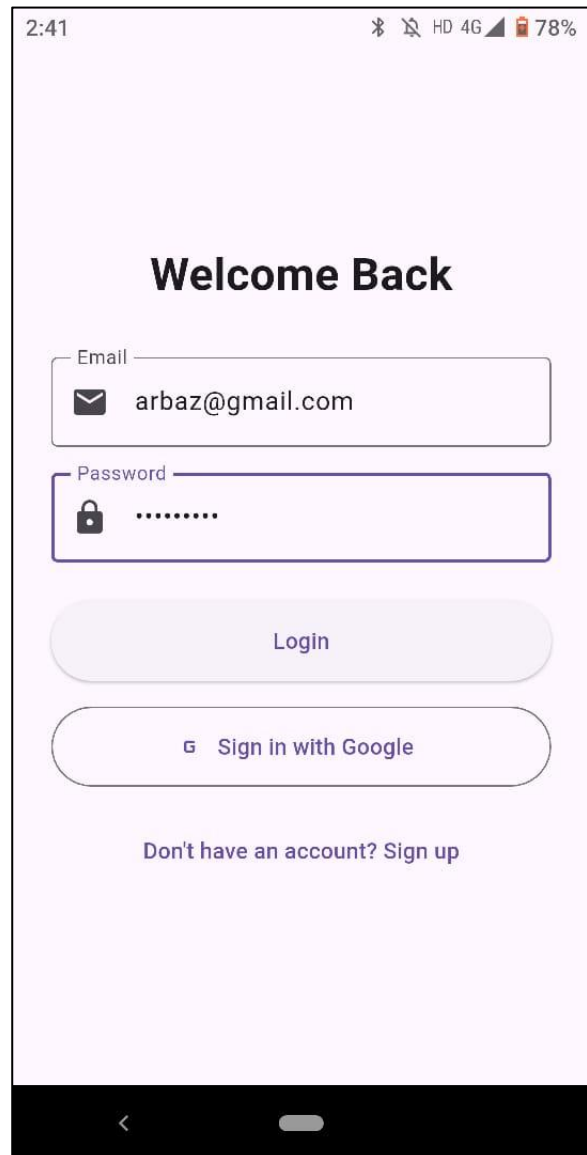
Email

Password

[Sign Up](#)

[Sign in with Google](#)

[Already have an account? Login](#)



2:41 HD 4G 78%

Welcome Back

Email

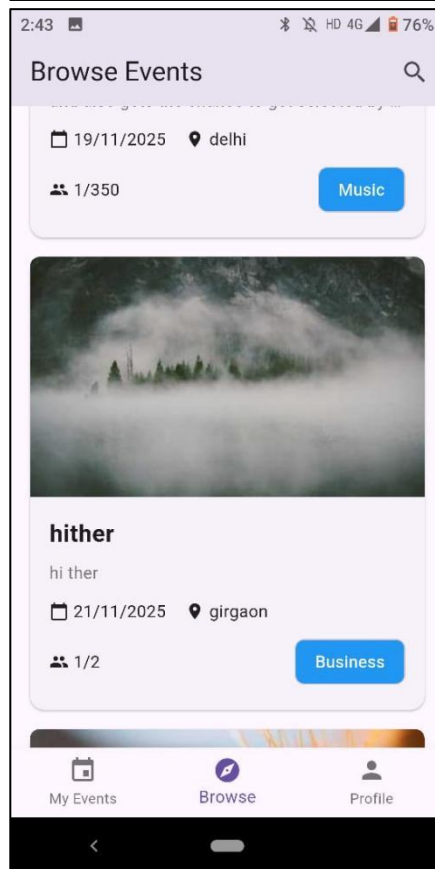
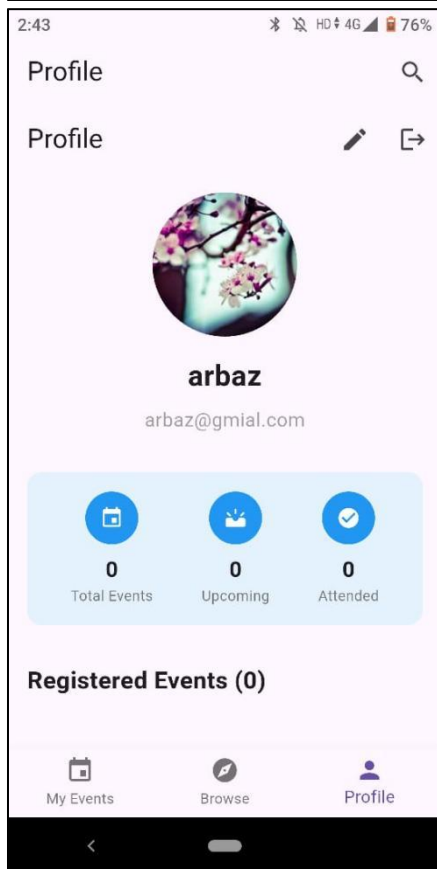
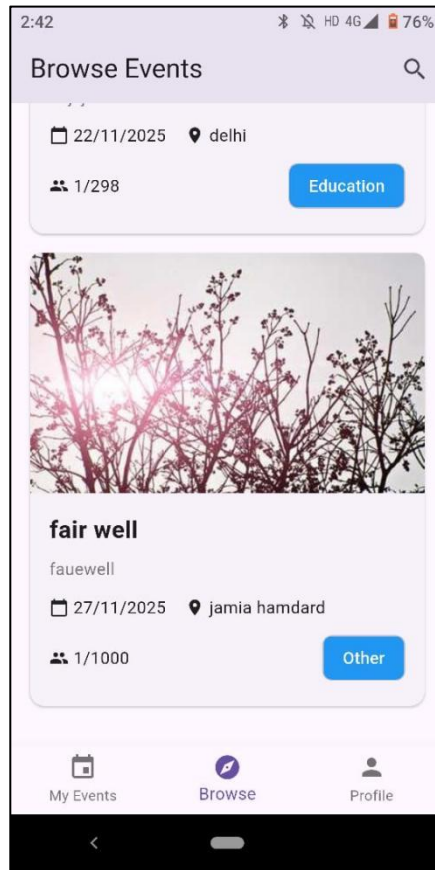
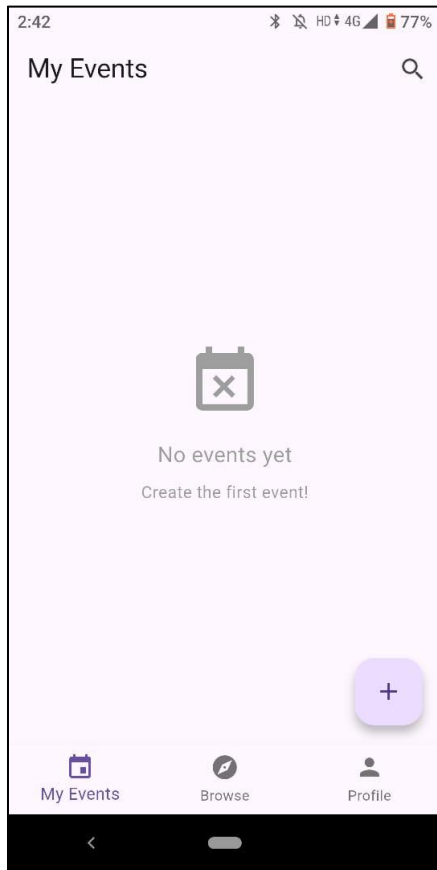
Password

[Login](#)

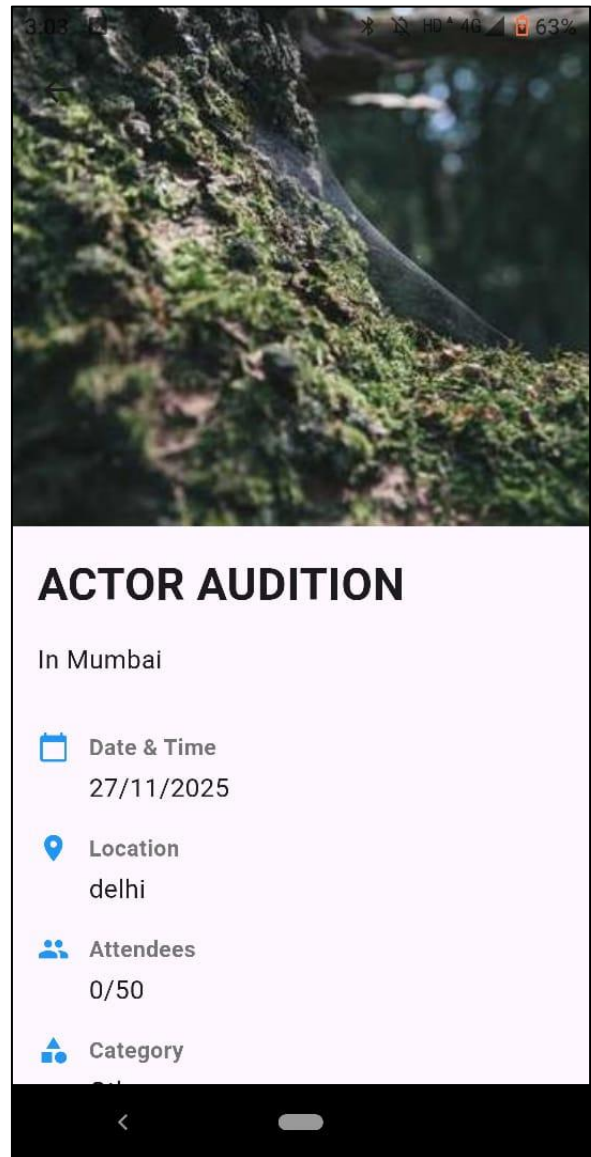
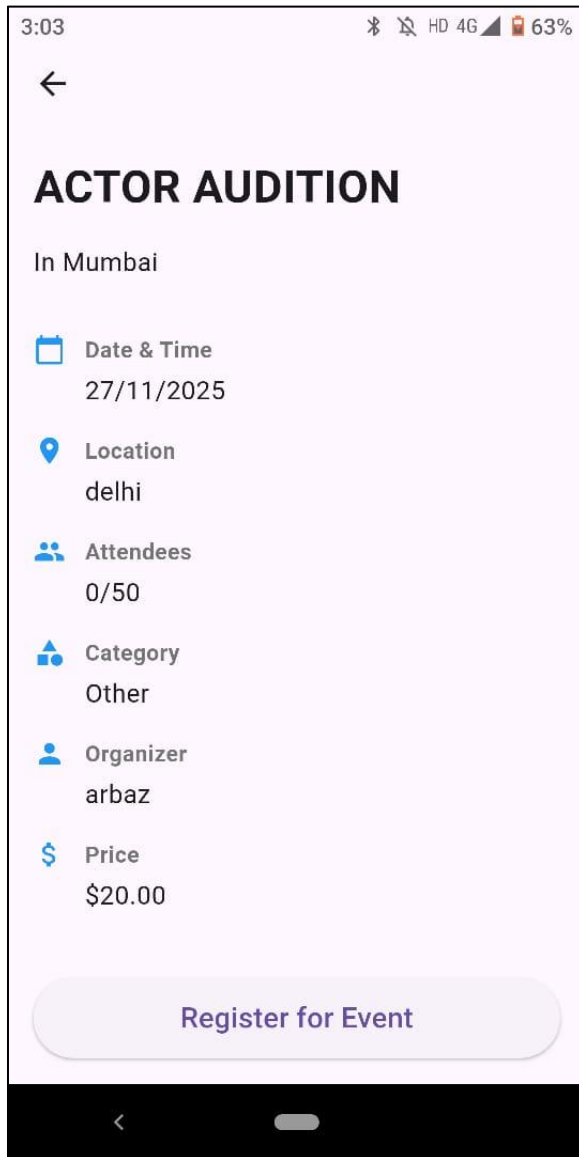
[Sign in with Google](#)

[Don't have an account? Sign up](#)

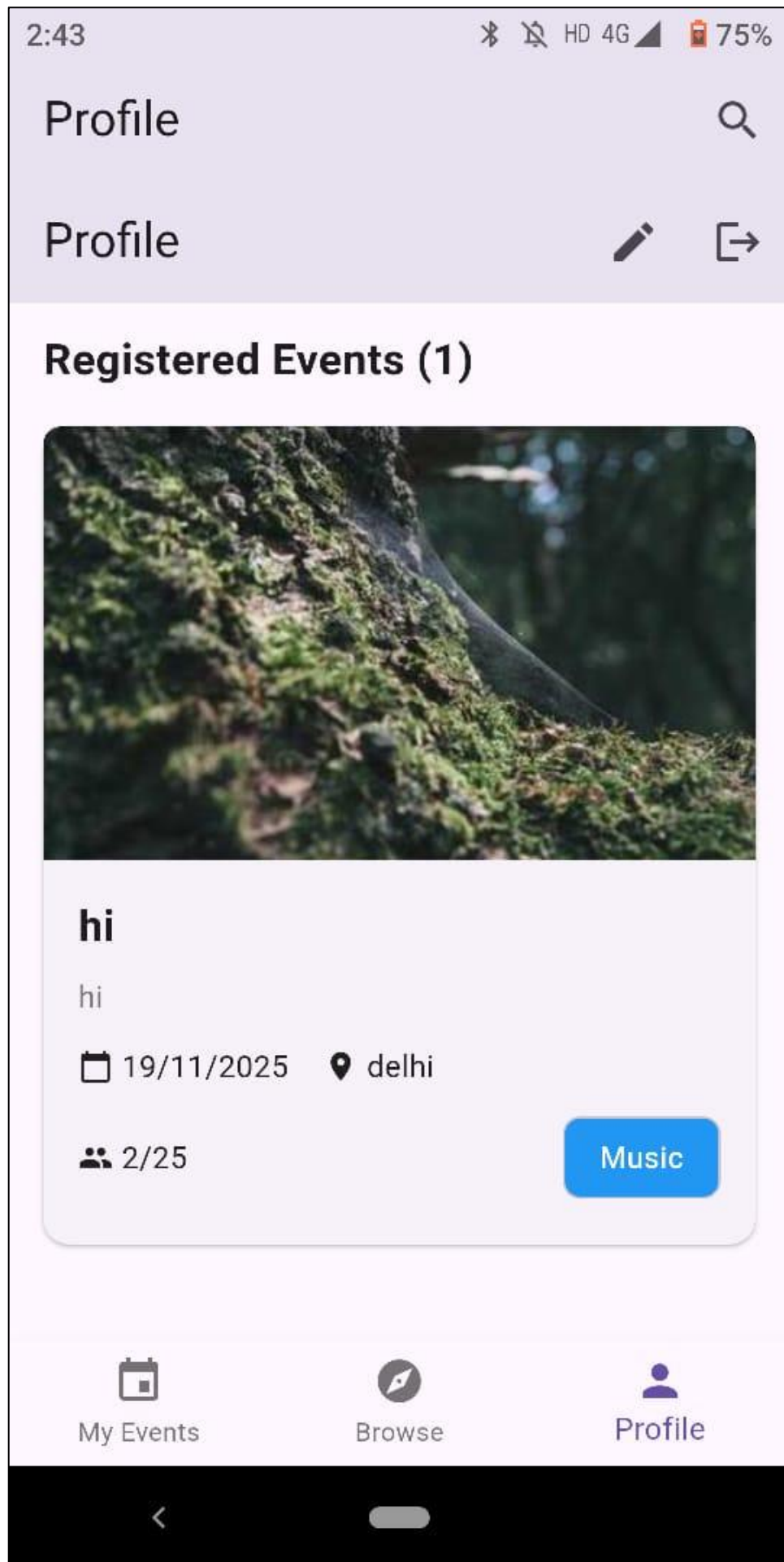
2. Main Screen (My Events, Browser Events, Profile):



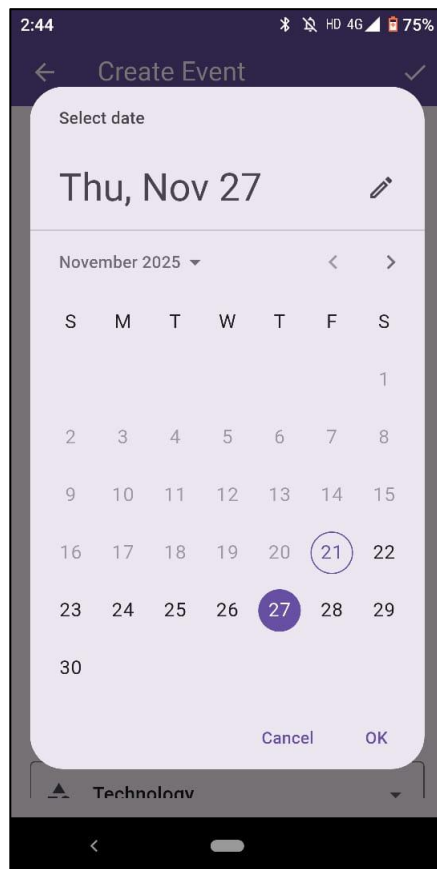
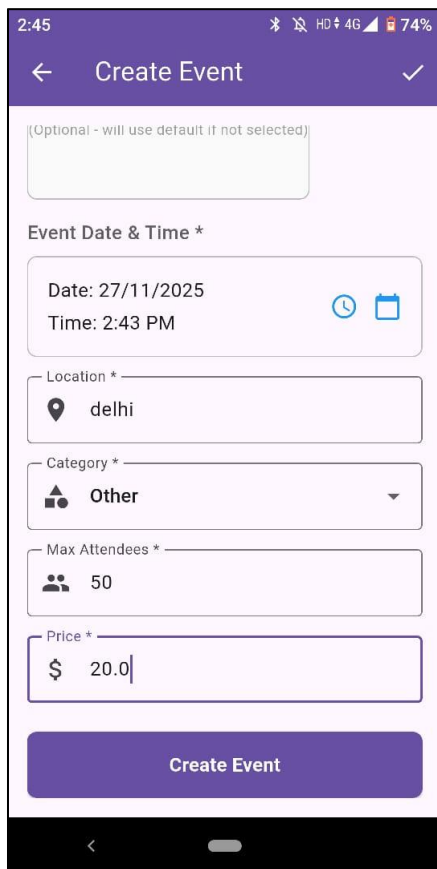
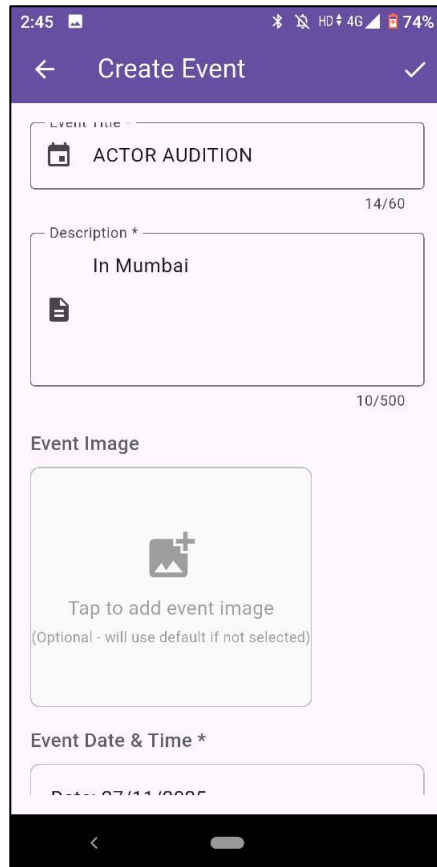
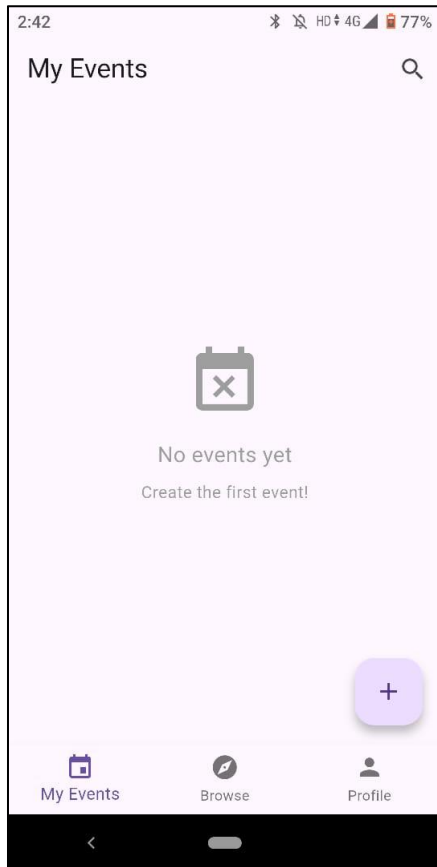
3. Registering Events(Title, Description, Date&Time, Location, Attendees, Category, Organizer, Price, Register for Event Button):

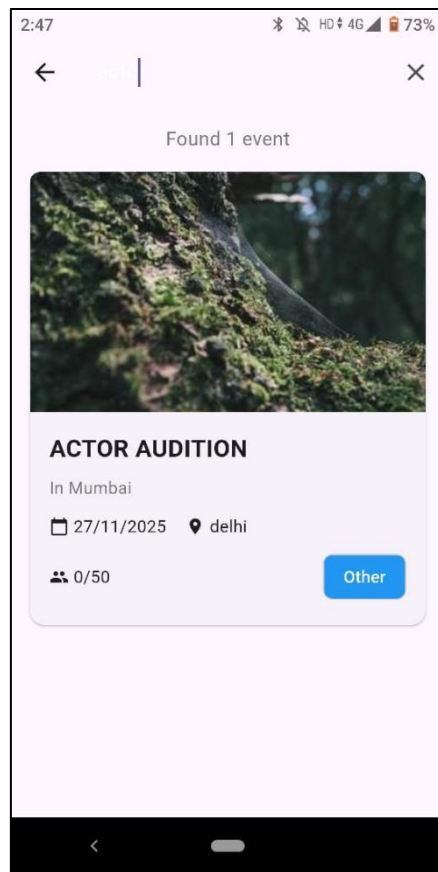
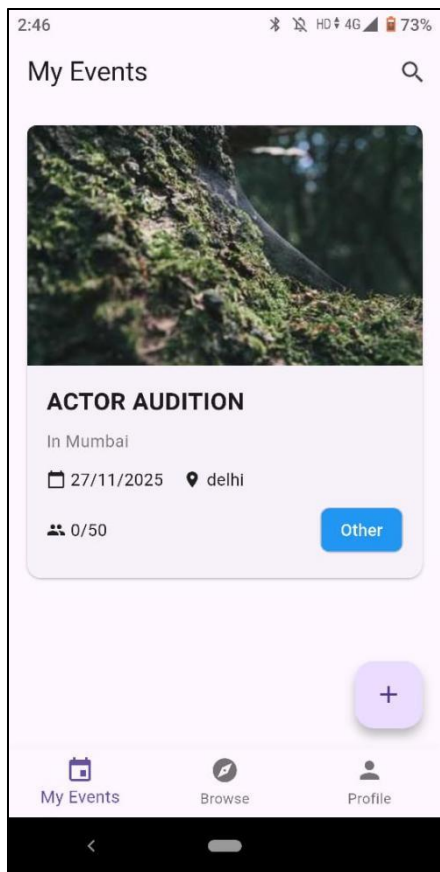


4. Registered Events:



5. Creating Events & Searching Events:





6. Internal Authentication:

Master Flutter Firebase Application (10) WhatsApp

console.firebase.google.com/u/0/project/event-management-59749/authentication/users

Event Management - Authentication

+

Event Management

Authentication

Users Sign-in method Templates Usage Settings Extensions

The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number, or user UID

Add user

Identifier	Providers	Signed In	Created	User UID
arbaz@gmail.com	☒	Nov 21, 2025	Nov 21, 2025	LC0Qcric4HdCMjwTIE9I9grS...
shakib@gmail.com	☒	Nov 21, 2025	Nov 21, 2025	uFebrezYXJZchHrAn74HrigCu...
anasansari@gmail.com	☒	Nov 20, 2025	Nov 20, 2025	4ts0TUSIEhRwm2UEZOdR5IK...
jai@gmail.com	☒	Nov 19, 2025	Nov 19, 2025	T63jUYuaTuUOF2fxNlHpgG9v...
gijas@gmail.com	☒	Nov 19, 2025	Nov 19, 2025	slrf5JQQTxh8rEW8WvE7to1YA...
	☒	Nov 19, 2025	Nov 19, 2025	jFsmjOyLoGbEMDeHtYsFe5m...

Project Overview

Firestore Database

Authentication

Realtime Database

Build

Run

Analytics

Related development tools

Firebase Studio

Spark

No-cost (\$0/month)

Upgrade

Type here to search

ENG 3:20 PM 11/21/2025

7. Firebase Firestore Database:

The screenshot displays the Firebase console interface. On the left, the sidebar includes a 'Firestore Database' button. The main content area is titled 'Event Management - Database' and shows a tree view of the database structure. The 'users' collection is expanded, revealing a document with the following fields:

- createdAt:** November 21, 2025 at 2:41:35 PM UTC+5:30
- email:** "arbaz@gmail.com"
- name:** "arbaz"
- profileImage:** "https://picsum.photos/seed/463/200/200"
- registeredEvents:** 0

The bottom of the screen shows a navigation bar with various Firebase services and a search bar.

Master Flutter Firebase Application

console.firebase.google.com/u/0/project/event-management-59749/firestore/databases/-default-/data/~2FvfyhQt5t5fY04n8mEWEL

Event Management - Database

+

Event Management

Cloud Firestore > Database

Project Overview

Firestore Database

Authentication

Realtime Database

Product categories

Build

Run

Analytics

Related development tools

Firestore Studio

Spark

No-cost (\$0/month)

Upgrade

events

+ Start collection

events

users

events

+ Add document

5XEXNcHfXGw6rLMhjZk

Fdb9NVGLFT1MGmOAUzPF

ddwGsv8d2KQbJ3koIOAZ

gjRPl0Vm3h1ewbPQX3o

jB2Y0aXufyzbIMdZKjHQ

kr6E9NclSmpX4sC3Y0uF

vPfyhQt5t5fY04n8mEWEL

wCRLmHb5bWtdRF6uWNpr

vPfyhQt5t5fY04n8mEWEL

+ Start collection

+ Add field

attendees

category: "Other"

createdAt: November 21, 2025 at 2:46:02 PM UTC+5:30

dateTime: November 27, 2025 at 2:43:00 PM UTC+5:30

description: "In Mumbai"

imageUrl: "https://picsum.photos/seed/978/400/300"

location: "delhi"

maxAttendees: 50

organizer: "arbaz"

organizerId: "LOOQcnie4HdCMjwTlE9I9grSxqE2"

price: 20

title: "ACTOR AUDITION"

More in Google Cloud

Type here to search

25°C Sunny

3:23 PM

ENG

IN

11/21/2025

Conclusion

- **Successfully Developed** a fully functional Event Management System mobile application using Flutter and Firebase
- **Achieved Cross-Platform Compatibility** with native performance on both iOS and Android devices
- **Implemented Core Features** including user authentication, event creation, and real-time event browsing
- **Established Scalable Architecture** using Firebase Firestore for data management and Firebase Storage for media handling
- **Created User-Centric Design** with intuitive navigation and Material Design principles
- **Demonstrated Effective Integration of multiple Firebase services** working seamlessly together
- **Delivered MVP (Minimum Viable Product)** that addresses the primary pain points of event organizers and attendees
- **Provided Solid Foundation** for future enhancements and feature additions
- **Ensured Data Security** through Firebase Authentication and security rules
- **Met Project Objectives** of creating a centralized platform for event management and discovery

Limitations

➤ **Functional Limitations:**

1. **No Event Modification** - Created events cannot be edited or deleted once published
2. **No RSVP System** - Users cannot indicate attendance or register for events
3. **Limited Search & Filtering** - Basic browsing without categories, tags, or advanced filters
4. **No User Profiles** - Limited user information and no personalization options
5. **No Notifications** - Absence of push notifications for new or updated events

➤ **Technical Limitations:**

1. **Mandatory Online Requirement** - Complete dependency on internet connectivity
2. **No Offline Support** - Application becomes non-functional without network connection
3. **Limited Error Handling** - Basic error messages without detailed user guidance
4. **No Data Caching** - Events need to be reloaded every time app is opened
5. **Single Image Upload** - Only one image per event with no gallery support

➤ **Platform Limitations:**

1. **Firebase Vendor Lock-in** - Heavy dependency on Google's Firebase ecosystem
2. **No Web Version** - Mobile-only application without web accessibility
3. **Limited Internationalization** - English language support only
4. **No Admin Panel** - Lack of administrative controls and moderation features
5. **Scalability Concerns** - Potential performance issues with very large event datasets

➤ **User Experience Limitations:**

1. **No Social Features** - Unable to share events or interact with other users
2. **Limited Event Discovery** - No personalized recommendations or trending events
3. **No Rating System** - Absence of event reviews and organizer ratings
4. **Basic Navigation** - Limited browsing options and sorting capabilities
5. **No Booking Management** - Cannot track event attendance or manage participants

Future Scope

- **Immediate Enhancements:**
 - **Event Editing & Deletion** - Allow organizers to modify or remove their events
 - **RSVP & Attendance System** - Enable users to register and confirm event participation
 - **Advanced Search & Filters** - Implement categories, tags, and location-based filtering
 - **Push Notifications** - Send alerts for new events, reminders, and updates
 - **User Profiles** - Create detailed user profiles with avatars and preferences
- **Medium-Term Features:**
 - **Payment Integration** - Support paid events with secure payment processing
 - **Offline Functionality** - Enable basic browsing without internet connection
 - **Social Features** - Add event sharing, reviews, and user interactions
 - **Multi-language Support** - Expand to regional languages for wider accessibility
 - **Web Version** - Develop responsive web application using Flutter Web
- **Advanced Capabilities:**
 - **Analytics Dashboard** - Provide organizers with event insights and attendance reports
 - **AI Recommendations** - Implement smart event suggestions based on user behavior
 - **Live Streaming** - Support virtual events with integrated video streaming
 - **Calendar Integration** - Sync events with Google Calendar and Apple Calendar
 - **Admin Panel** - Develop moderation tools and content management system
- **Platform Expansion:**
 - **Tablet & Desktop Support** - Optimize for larger screen devices
 - **API Development** - Create public APIs for third-party integrations
 - **International Expansion** - Support multiple currencies and time zones
 - **Enterprise Features** - Add team management and bulk event operations

Bibliography

- <https://docs.flutter.dev/ui/widgets/scrolling>
- <https://docs.flutter.dev/ui/widgets/material>
- <https://api.flutter.dev/flutter/material/NavigationBar>
- <https://api.flutter.dev/flutter/material/FloatingActionButton/FloatingActionButton>
- https://www.youtube.com/watch?v=2HaKjVXyGbs&list=PLWNYjcx_ZHPeamRJbGhn8pH1AmmCw48F0
- https://www.youtube.com/watch?v=uOWeCc_e1RY
- <https://api.flutter.dev/flutter/material/showDatePicker>
- <https://docs.flutter.dev/>
- https://www.youtube.com/watch?v=4Hi_7PvpGSA&t=6s